Lab - 12

Apply the K-mean Clustering on the following data and form the Cluster for the value of K = 2.

Sr.	X	Y
1	1	1
2	1.5	2
3	3	4
4	5	7
5	3.5	5
6	4.5	5
7	3.5	4.5

→ Here K = 2 and we assume Center is (1,1) for cluster 1 and (5,7) for cluster 2.

Data Points		Distance to Center		Cluster
X	Y	1, 1	5, 7	
1	1	0	7.21	1
1.5	2	1.11	6.1	1
3	4	3.6	3.6	1
5	7	7.2	0	2
3.5	5	4.7	2.5	2
4.5	5	5.3	2.06	2
3.5	4.5	3	2.9	2

Euclidean Distance Formula = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

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New Centroid:-

$K_1 = (1.83, 2.33)$
$K_2 = (4.12, 5.37)$

Data points		Distance to center		Cluster	New Cluster
X	Y	1.83, 2.33	4.12, 5.37		
1	1	1.57	5.4	1	1
1.5	2	0.46	4.22	1	1
3	4	2.03	1.77	1	2
5	7	5.64	1.85	2	2
3.5	5	3.14	0.72	2	2
4.5	5	3.77	0.53	2	2
3.5	4.5	2.74	1.07	2	2

New Centroid:-

$K_1 = (1.25, 1.5)$
$K_2 = (3.9, 5.1)$

Data points		Distance to centers		Cluster	New cluster
X	Y	1.25, 1.5	3.9, 5.1		
1	1	0.55	5.02	1	1
1.5	2	0.56	3.92	1	1
3	4	3.05	1.42	2	2
5	7	6.66	2.19	2	2
3.5	5	4.15	0.14	2	2
4.5	5	4.77	0.6	2	2
3.5	4.5	3.75	0.72	2	2

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Here we will stop as there is no change in the clusters.

Final clusters :- 1 → {1, 2}
2 → {3, 4, 5, 6, 7}